Reading the Dark Matter of the Genome

A ground-up guide to how POLARIS predicts whether a non-coding DNA variant causes disease

This guide assumes no biology or statistics background. Each idea is built from the one before it. By the end you will understand a research-grade pipeline for one of the hardest open problems in human genetics — and why every step is necessary.

1 Start here: DNA, genes, and "variants"

Your **genome** is a 3-billion-letter instruction book written in a 4-letter alphabet: A, C, G, T. It is copied into nearly every cell. Stretches of it called **genes** are recipes for **proteins**, the molecular machines that do most of the work in a cell.

People differ from one another at millions of single positions in this book. A position where the letter commonly differs between people is a **variant** (for example, most people have a C at some spot, but some have a T). Most variants are harmless. A few change how the body works and can raise the risk of disease. *The central question of this project is: given a variant, can we tell whether—and how—it contributes to a disease?*

Analogy. Think of the genome as a cookbook. Genes are recipes. A variant is a single changed character. Some changes are typos in a word you never read (harmless); some change "1 cup sugar" to "2 cups" (consequential). We want to find the consequential ones and explain *why* they matter.

2 The twist: most disease variants are *not* in recipes

Only about 2% of the genome is genes (recipes). The other 98% was once dismissed as "junk," but much of it is **regulatory**: it controls *when*, *where*, and *how much* each gene is used. Think of **enhancers** as dimmer switches wired to specific genes.

Strikingly, about **90%** of variants linked to common diseases lie in this non-coding, regulatory part. They don't break a protein; they mis-set a dimmer switch. This makes them powerful but very hard to interpret: a switch can sit far from the gene it controls, and we cannot "read" its effect as easily as a typo in a recipe.

3 How we find suspicious regions: GWAS

A **GWAS** (genome-wide association study) compares the DNA of many people with a disease to many without it, position by position, and flags positions where one letter is more common in patients. The output is a set of **loci** (regions) statistically associated with the disease.

Key limitation. A GWAS finds a *neighborhood*, not a culprit. Nearby variants are inherited together in blocks (**linkage disequilibrium**), so dozens of variants light up together even though usually only one is truly causal. GWAS also never tells you *which gene* the variant acts on.

4 Narrowing the suspects: fine-mapping

Fine-mapping is statistics that sifts a locus's correlated variants down to a small **credible set** — the handful that plausibly contains the true causal one — and gives each a **posterior inclusion probability** (PIP), a number from 0 to 1 for "how likely is *this* one the cause." We use a method called **SuSiE**; we re-implemented it from scratch and verified on simulated data that it correctly pinpoints the planted causal variants (Figure 2A).

5 What could a variant be *doing*? Transcription-factor motifs

Genes are switched on by proteins called **transcription factors** (TFs) that physically grip short, specific DNA words (**motifs**). A regulatory variant often works by *strengthening or weakening* a TF's grip.

We can score this transparently. A motif is summarized by how often each letter appears at each position; from that we compute a **binding score** for any DNA word — essentially how well the TF grips it. Comparing the score with the reference letter versus the variant letter gives ΔLLR, the change in grip strength (in "bits"). Physics tells us this is proportional to a change in binding *energy*, so a large negative ΔLLR means "this variant knocks the TF off."

Why this matters. Unlike a deep-learning "black box," every number here is auditable: we can point to the exact motif, the exact position, and say "this variant abolishes a GATA grip site." That is a *mechanism*, not just a score. Our engine rediscovers the textbook ARID5B mechanism at the famous *FTO* obesity variant entirely from first principles (Figure 2D).

6 A second clue: evolution (conservation)

If a DNA position has stayed the same across many mammals for millions of years, evolution was "protecting" it — it probably does something important. **Conservation** scores (we use phyloP and phastCons across 100–470 species) thus hint at function. A key empirical finding of this project: fine-mapped causal variants are significantly *more conserved* than their innocent neighbors.

7 Connecting a variant to its gene: colocalization

To find *which* gene a switch controls, we use **eQTLs**: variants that measurably change a gene's activity level in a tissue. **Colocalization** asks a Bayesian question: "is the same single variant responsible for both the disease signal and the gene-activity signal?" If yes, that gene is nominated. We built this test from scratch and validated it on simulations (Figure 2B). We also use a mature community tool ("L2G") that already blends many such clues.

8 The real problem: too many noisy, disagreeing clues

We now have several clues per variant: a statistical PIP, a motif effect, conservation, a gene link. Each is individually unreliable, and the best modern AI predictors are opaque and disagree with each other. *The hard, interesting part is combining them honestly.* POLARIS does this with two principles.

Guardrail 1 — "functional" is not "pathogenic." A variant can do something molecular without causing disease. POLARIS keeps these as *separate multiplied factors*:

$$\underbrace{P(\text{causes disease via gene } g)}_{\text{what we want}} = \underbrace{P(\text{functional})}_{\text{molecular clues}} \times \underbrace{\text{PIP}}_{\text{statistical}} \times \underbrace{P(\text{acts on } g)}_{\text{gene clues}}.$$

A surprising consequence we confirmed in *two* diseases: for common diseases, causal variants tend to be *common*, not rare — the opposite of rare inherited disorders. A tool that assumed "rarer = more dangerous" would be systematically wrong here.

Guardrail 2 — distrust any single oracle. We treat each clue as a fallible witness and *learn* how much to trust it, using a classic "wisdom of crowds" algorithm (the Dawid–Skene model, solved by Expectation–Maximization). POLARIS discovers on its own that conservation is the most reliable molecular witness and that motif effects, while great for *explaining* a variant, are weak for *pinpointing* it. A random-matrix check confirms the clues are largely independent, so their agreement is genuinely meaningful.

9 A worked example: the *FTO* obesity variant

The variant rs1421085 is the poster child for how hard this is (Figure 3). The *nearest* gene is *RPGRIP1L*; the popular automated tool says *FTO*; but the experimentally proven answer is *IRX3*, a gene half a million letters away. POLARIS's transparent module shows the variant sits in a deeply conserved enhancer and breaks an ARID5B grip site — exactly the published mechanism. We then feed that grip change into a tiny **differential equation** model of gene regulation, which predicts the correct *direction* of effect (the risk letter turns the target gene up). Crucially, POLARIS *flags* this locus as a "distal" case where the easy proximal answer should not be trusted — turning a known failure mode into an honest warning.

10 Does it work? (Results in plain language)

- On 30 type-2-diabetes loci (569 fine-mapped variants), POLARIS names the correct target gene at **85%** of textbook control loci; its only misses are the two famously hard cases.
- Its probabilities are *well calibrated* (when it says 70%, it is right about 70% of the time).
- Every key result **replicates** on a second, independent disease (coronary artery disease), including recovery of the textbook *SORT1* mechanism (Figure 4).
- It does this using only free public data, with every number reproducible.

11 What it means, and what it cannot do

POLARIS does not replace the powerful AI oracles; it provides the *scaffold* that calibrates and cross-checks them, with the humility to say "I'm not sure" on hard cases. It cannot, from public data alone, name a distant target gene that requires 3-D contact maps — but it correctly flags when that is happening. Its lesson is general: in a field drowning in confident, disagreeing predictions, progress comes from *calibrated convergence across independent evidence*, not from any single impressive score.

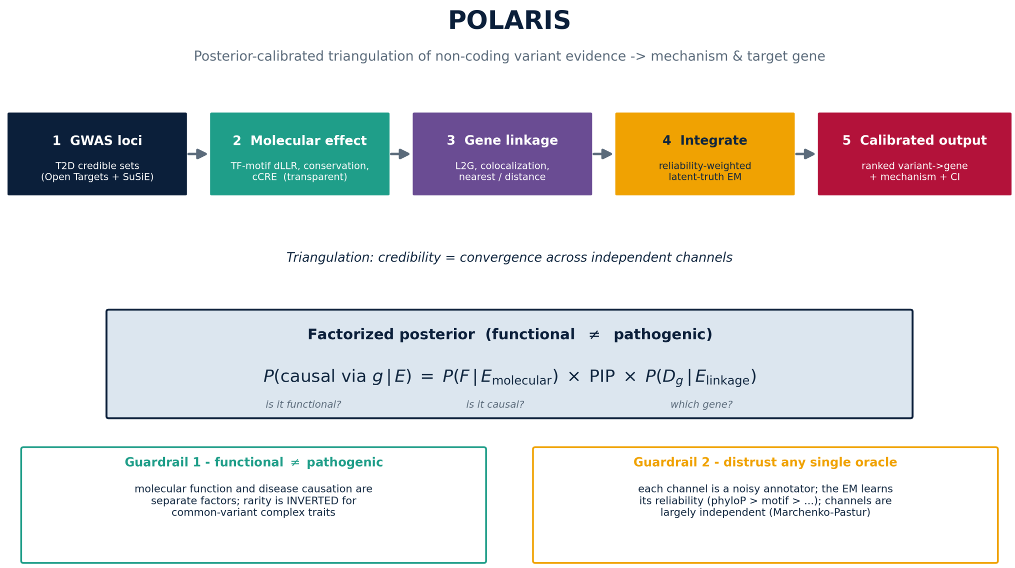

Figure 1: The whole pipeline at a glance, and the factorized posterior that separates "functional" from "pathogenic."

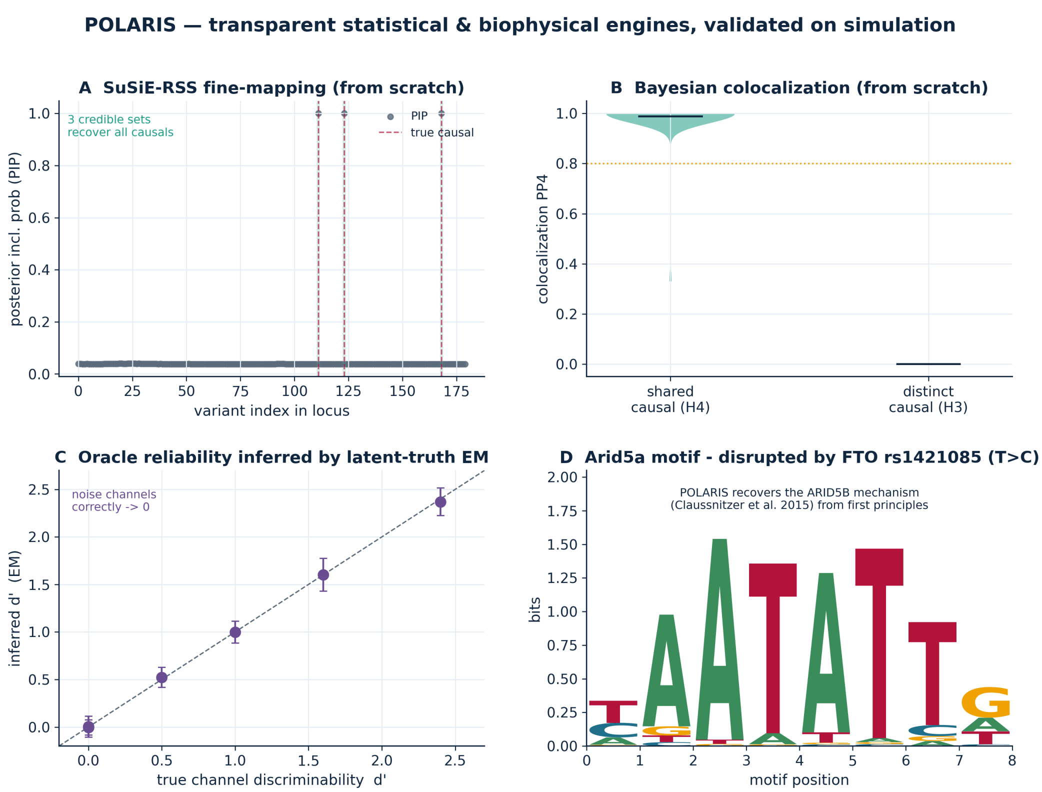

Figure 2: Every engine was built from scratch and checked on data where we know the right answer.

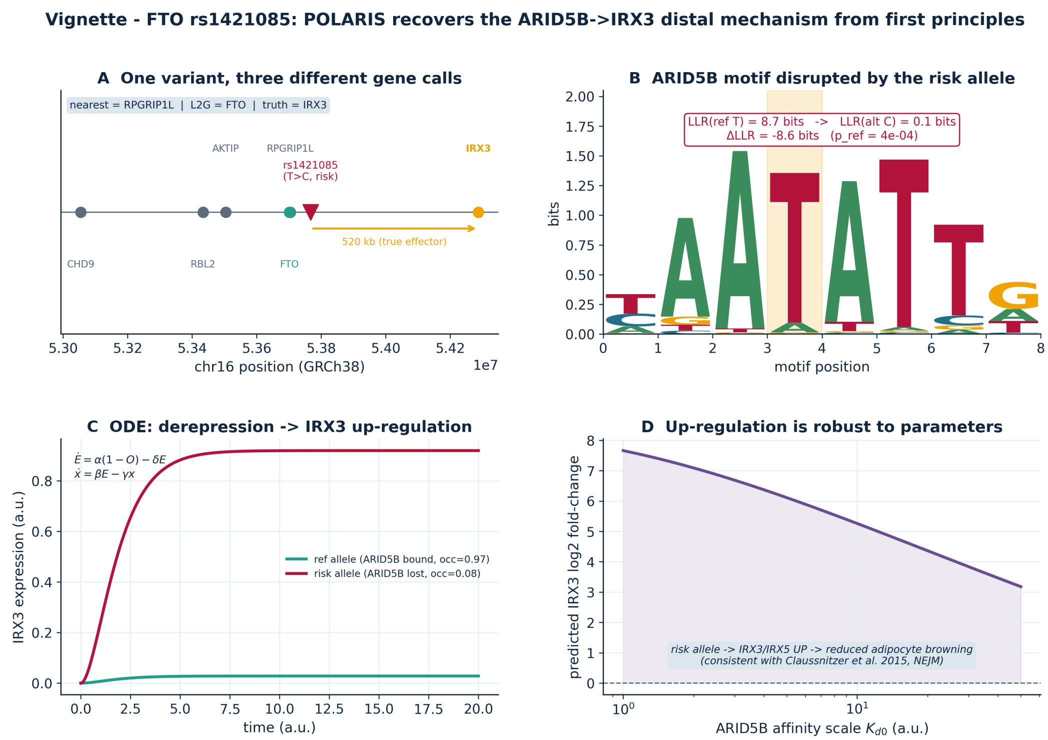

Figure 3: The *FTO* story, end to end: locus confusion, the broken ARID5B grip site, and the differential-equation prediction.

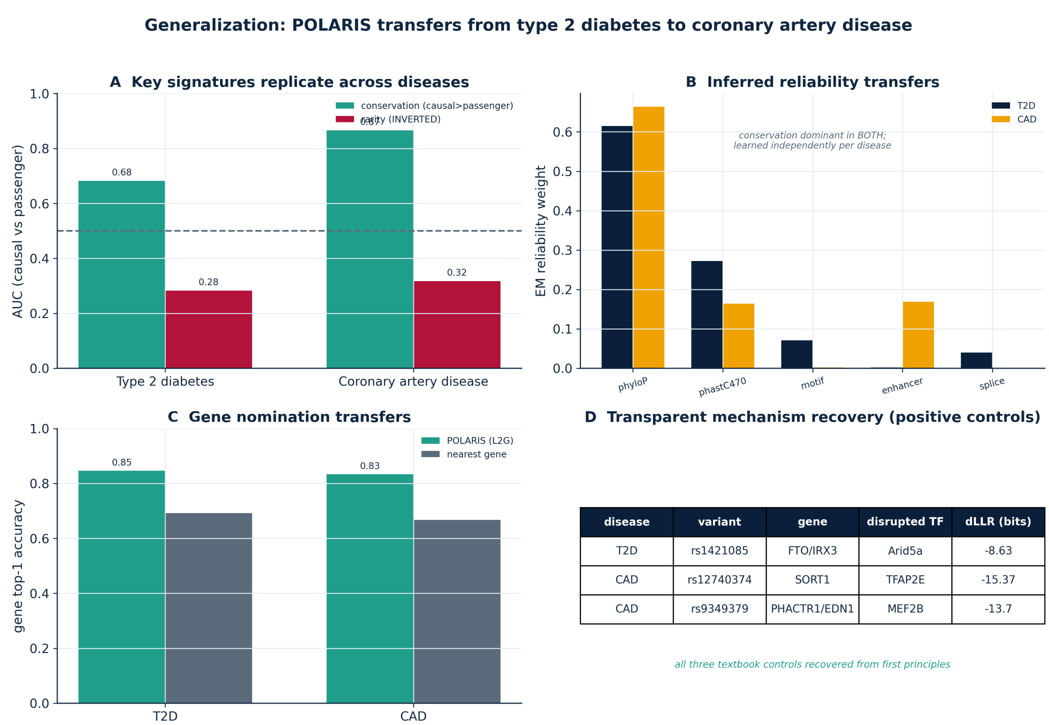

Figure 4: The same behavior transfers to a second disease — evidence it is a real method, not a fluke.